

Functional Annotation of *frmC* (PP_1617; UniProt Q88MF4) in *Pseudomonas putida* KT2440

Summary

frmC (locus tag PP_1617; UniProt Q88MF4) of *Pseudomonas putida* KT2440 encodes S-formylglutathione hydrolase (SFGH; EC 3.1.2.12), a cytosolic serine thioesterase/esterase of the esterase D (ESD) / α/β -hydrolase-fold family. Its primary catalytic function is the hydrolysis of S-formylglutathione to formate and reduced glutathione (GSH): **S-formylglutathione + H₂O → formate + GSH**. This is the terminal, glutathione-regenerating step of the near-universal glutathione (GSH)-dependent formaldehyde-detoxification pathway. By releasing formate and returning GSH to the free thiol pool, FrmC completes a catalytic cycle in which GSH is used as a disposable C1-carrier cofactor rather than being consumed.

The gene's identity is unambiguous and well supported. The 284-residue protein carries the diagnostic hallmarks of this enzyme class: a GxSxG nucleophile-elbow motif (G-H-**S150**-M-G) housing the catalytic serine, a UniProt-annotated Ser150–His261–Asp228 charge-relay triad on an α/β -hydrolase fold, and InterPro/TIGRFAM signatures (IPR014186 S-formylglutathione hydrolase; IPR000801 esterase-D-like; Pfam PF00756). FrmC shares 57–62% sequence identity with the two biochemically characterized *E. coli* SFGHs (FrmB and YeiG), with its catalytic serine and histidine aligning exactly to residues that alanine-scanning mutagenesis proved to be essential. Genomically, *frmC* (PP_1617) lies immediately downstream of and co-oriented with *frmA* (PP_1616), the GSH-dependent formaldehyde dehydrogenase, separated by only 8 bp — a two-gene formaldehyde-detoxification operon that mirrors the conserved *fghA/flhA* arrangement in *Paracoccus denitrificans*.

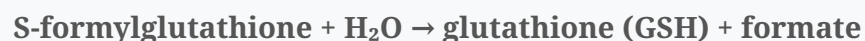
Functionally, FrmC operates in the **cytoplasm** (no signal peptide or transmembrane segment; GO:0005829) and is the **sole SFGH** encoded in the KT2440 genome, making it non-redundant. Its physiological importance is heightened in this organism because *P. putida* KT2440 catabolizes lignin-derived methoxylated aromatics (e.g., vanillate), whose O-demethylation by Rieske monooxygenases liberates cytotoxic formaldehyde. FrmC clears this formaldehyde via the GSH pathway; consistent with this, adaptive laboratory evolution for improved vanillate

utilization selected a mutation in the formaldehyde-detoxification (fghA/SFGH) step. The enzyme's substrate specificity — driven by a narrow acyl-binding pocket — favors S-formylglutathione and other small-acyl thioesters, with retained promiscuous short-chain carboxylesterase activity typical of the esterase D family, whose deeply conserved human ortholog is esterase D (ESD).

Key Findings

Finding 1 — frmC is S-formylglutathione hydrolase catalyzing the final step of GSH-dependent formaldehyde detoxification

UniProt Q88MF4 (frmC / PP_1617, *P. putida* KT2440, 284 aa) is annotated as **S-formylglutathione hydrolase**, EC 3.1.2.12 (NCBIfam TIGR02821; RuleBase RU363068). The catalyzed reaction is:



Gene Ontology assignments consistently describe this activity: S-formylglutathione hydrolase activity (GO:0018738), carboxylic ester hydrolase activity (GO:0052689), and formaldehyde catabolic process (GO:0046294). InterPro (IPR014186, S-formylglutathione_hydrol; IPR000801, esterase-like/esterase D) and Pfam (PF00756) domain assignments corroborate the family placement.

This function is not merely a database inference: multiple orthologs have been biochemically characterized and shown to prefer S-formylglutathione as physiological substrate. In *E. coli*, the two SFGHs FrmB and YeiG "had the highest hydrolytic activity toward S-formylglutathione, an intermediate of the glutathione-dependent pathway of formaldehyde detoxification," with a similar affinity ($K_m = 0.41\text{--}0.43$ mM) [PMID: 16567800]. The *Paracoccus denitrificans* ortholog FghA is part of "a formaldehyde detoxification pathway that is universal in nature" [PMID: 8892832]. The reaction and pathway position are stated directly for the plant ortholog: the enzyme "catalyses the final step of formaldehyde detoxification: the hydrolysis of S-formylglutathione to formate and glutathione" [PMID: 14646092]. Together these establish that FrmC performs the terminal, GSH-regenerating step of the pathway.

Finding 2 — FrmC is a serine hydrolase with an α/β -hydrolase fold and a Ser150–His261–Asp228 catalytic triad

Sequence analysis of Q88MF4 identifies a **nucleophile-elbow motif G-H-S150-M-G** — the canonical GxSxG signature — placing **Ser150 as the catalytic nucleophile**. UniProt annotates a charge-relay system comprising **Ser150, Asp228, and His261**. The fold and family assignments are mutually consistent: SUPFAM SSF53474 (α/β -hydrolases), Pfam PF00756 (Esterase), InterPro IPR029058 (AB_hydrolase_fold) and IPR000801 (esterase-D-like); UniProt keyword "Serine esterase."

The catalytic mechanism is experimentally validated in orthologs. In *E. coli* YeiG, "Ser145, Asp233, and His256 are absolutely required for activity, indicating that these residues represent a serine hydrolase catalytic triad in this protein and in other S-formylglutathione hydrolases" [PMID: 16567800]. Structural and biochemical work on the Arabidopsis enzyme confirms that "AtSFGH from Arabidopsis thaliana has been shown to be a serine hydrolase rather than a cysteine hydrolase" [PMID: 16626737]. Notably, a conserved reactive cysteine near the active site (Cys57 in FrmC; cf. Cys59 in AtSFGH) "lies in close proximity to the serine hydrolase triad, serving a gate-keeping function" [PMID: 16626737], suggesting a possible redox-regulatory role for FrmC's activity. Yeast crystallography (PDB 1pv1; Ser161/His276/Asp241) further supports the conserved triad geometry.

Finding 3 — FrmC has a narrow acyl-binding pocket giving specificity for S-formylglutathione and small acyl esters; it acts in the cytosol

Substrate specificity in this family is governed by the size of the acyl-binding pocket. The psychrophilic SFGH from *Pseudoalteromonas haloplanktis* (PhEst) "is able to efficiently hydrolyze several thioester substrates with very small acyl moieties. By contrast, the enzyme shows no activity toward substrates with bulky acyl groups... a very narrow acyl-binding pocket in a typical alpha/beta-hydrolase fold" [PMID: 20209484]. This explains the strong preference for the small **formyl** group of S-formylglutathione. Alongside the primary thioesterase activity, these enzymes retain promiscuous carboxylesterase activity: purified *E. coli* FrmB and YeiG "have high carboxylesterase activity against the model substrates, p-nitrophenyl esters of fatty acids (C2-C6) and alpha-naphthyl acetate" [PMID: 16567800], the classic esterase-D activity — but the highest activity is on S-formylglutathione.

Localization: Q88MF4 is a 284-aa soluble protein with no signal peptide and no transmembrane region; the GSH-dependent formaldehyde pathway operates in the bacterial cytoplasm, and the GO assignment is cytosol (GO:0005829). FrmC therefore carries out its function intracellularly, in the cytosol.

Finding 4 — frmC (PP_1617) forms a two-gene formaldehyde-detoxification operon with frmA (PP_1616)

KEGG genome analysis of *P. putida* KT2440 places the two genes consecutively on the same strand:

Locus	Gene	Function	KO / EC	Coordinates (strand)
PP_1616	frmA	GSH-dependent formaldehyde dehydrogenase / S-(hydroxymethyl)glutathione dehydrogenase (class III ADH)	K00121; EC 1.1.1.284 / 1.1.1.1	1,812,522–1,813,637 (+)
PP_1617	frmC	S-formylglutathione hydrolase	K01070; EC 3.1.2.12	1,813,646–1,814,500 (+)

The **8 bp intergenic gap** and shared orientation indicate co-transcription as an operon encoding two consecutive steps of the same pathway. Immediately adjacent is PP_1615, a LysR-family transcriptional regulator (complementary strand) — a candidate regulator. Flanking genes PP_1618 (MEP-pathway IspF) and PP_1619 (tRNA pseudouridine synthase) are functionally unrelated, delimiting the operon.

This arrangement is evolutionarily conserved: in *Paracoccus denitrificans*, "Downstream of flhA, the... gene encoding glutathione-dependent formaldehyde dehydrogenase, an open reading frame was identified and called fghA" [PMID: 8892832] — exactly the frmA→frmC layout. The operon is expected to be formaldehyde-inducible, as "the expression of frmB was stimulated 45-75 times by the addition of formaldehyde to the growth medium" in *E. coli* [PMID: 16567800].

Finding 5 — FrmC belongs to the deeply conserved esterase D family and completes a pathway whose upstream steps are spontaneous adduct formation followed by NAD⁺-dependent oxidation

SFGHs are ubiquitous across prokaryotes and eukaryotes; the human ortholog **esterase D (ESD)** is itself an S-formylglutathione hydrolase. Literature explicitly equates the two: "S-formylglutathione hydrolase (esterase D; ESD)" [PMID: 24682756], and "Esterase D (ESD) is a nonspecific esterase that detoxifies formaldehyde" [PMID: 34875997]. *Paracoccus* FghA is homologous to human esterase D [PMID: 8892832].

The complete pathway around FrmC has three steps:

1. **Spontaneous adduct formation** — "The first step in the reaction involves the nonenzymatic formation of the S-(hydroxymethyl)glutathione adduct from formaldehyde and glutathione" [PMID: 10978156].
2. **NAD⁺-dependent oxidation** — S-hydroxymethylglutathione (HMGS) is oxidized by the GSH-dependent formaldehyde dehydrogenase / class III ADH (FrmA / PP_1616; EC 1.1.1.284) to S-formylglutathione.
3. **Hydrolysis by FrmC** — S-formylglutathione → formate + GSH.

This delineates FrmC's precise enzymatic scope: it acts only on the third step; the first step is non-enzymatic, and the second belongs to its operonic partner FrmA.

Finding 6 — FrmC shares 57–62% identity with characterized *E. coli* SFGHs, with exact conservation of mutagenesis-validated catalytic residues

Global pairwise alignment of FrmC (Q88MF4, 284 aa) to characterized orthologs:

Ortholog	UniProt	Length	% Identity	Aligned columns
<i>E. coli</i> YeiG (SFGH2)	P33018	278 aa	61.7%	277
<i>E. coli</i> FrmB (SFGH1)	P51025	277 aa	56.7%	275

Critically, the catalytic residues map exactly: **FrmC Ser150 ↔ YeiG Ser145 / FrmB Ser145** (nucleophile); **FrmC His261 ↔ YeiG His256 / FrmB His254**; and FrmC Asp228 aligns to the conserved catalytic aspartate region. Because "Ser145, Asp233, and His256 are absolutely required for activity... in this protein and in other S-formylglutathione hydrolases" in YeiG by

alanine-scanning mutagenesis [PMID: 16567800], the functional evidence transfers to FrmC by orthology. The catalytic serine sits in the GHS150MG (GxSxG) nucleophile elbow, the structural hallmark of the family.

Finding 7 — In *P. putida* KT2440, *frmC* is the sole SFGH and clears formaldehyde released by aromatic O-demethylation

A UniProt taxonomy query (organism_id 160488) returns exactly **one** S-formylglutathione hydrolase — Q88MF4 = *frmC* / PP_1617 — with **no paralog**, in contrast to *E. coli*, which has two (*FrmB* and *YeiG*). *FrmC* is therefore non-redundant in KT2440.

The physiological relevance is amplified by this strain's lifestyle. *P. putida* KT2440 catabolizes lignin-derived methoxylated aromatics; O-demethylation of vanillate by the Rieske monooxygenase VanAB releases formaldehyde as a toxic byproduct. In adaptive laboratory evolution selecting improved vanillate utilization, one of three impactful mutations mapped to genes "including those in VanB, the reductase for vanillate O-demethylase, PP_3494, a global regulator of vanillate catabolism, and *fghA*, involved in formaldehyde detoxification" [PMID: 38936762]. This directly ties the SFGH step to the handling of formaldehyde generated during aromatic catabolism. In authoritative pathway terms, the GSH-dependent route is one of several cofactor-dependent linear formaldehyde-oxidation pathways involving "the spontaneous or enzyme-catalyzed condensation of formaldehyde and the respective C(1) carrier, the oxidation of the cofactor-bound C(1) unit and its conversion to formate, and the oxidation of formate to CO(2)" [PMID: 12209256].

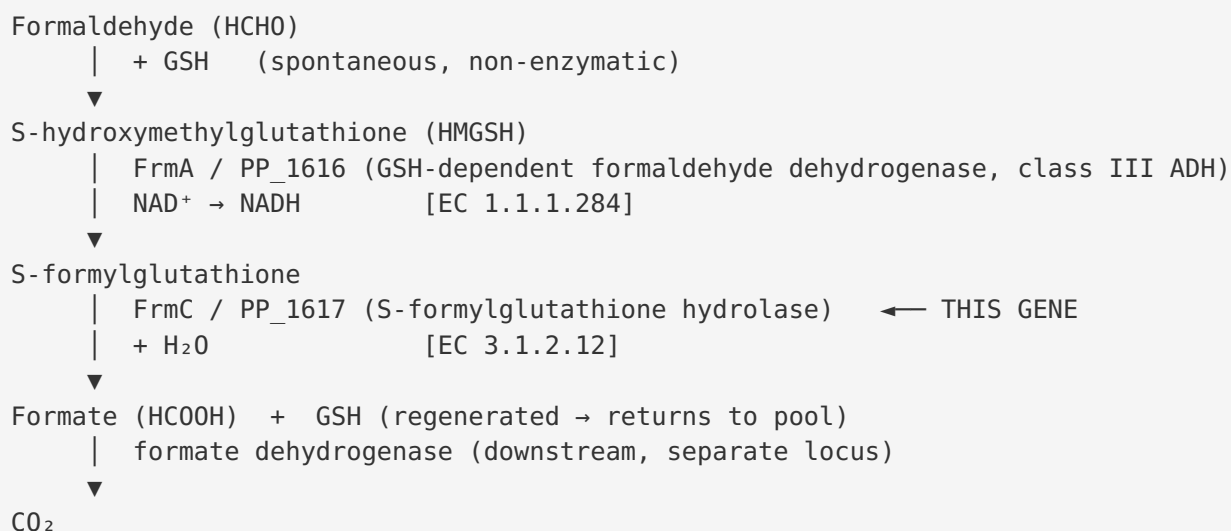
Finding 8 — Synthesis

Convergent evidence across all findings establishes *frmC*/PP_1617 as a **cytosolic serine S-formylglutathione hydrolase performing the terminal, GSH-regenerating step of glutathione-dependent formaldehyde detoxification** in *P. putida*: (1) UniProt/InterPro/TIGRFAM annotation as SFGH, EC 3.1.2.12; (2) Ser150 in a GxSxG elbow with an annotated Ser150–His261–Asp228 charge-relay triad; (3) 57–62% identity to *E. coli* SFGHs with exact catalytic-residue conservation; (4) a narrow acyl pocket conferring S-formylglutathione specificity plus secondary short-chain carboxylesterase activity; (5) cytosolic localization; (6) the *frmA*–*frmC* operon (8 bp gap); (7) sole-SFGH, non-redundant status clearing formaldehyde from aromatic O-demethylation; and (8) deep conservation to human esterase D.

Mechanistic Model / Interpretation

The glutathione-dependent formaldehyde-detoxification cycle

Formaldehyde is a reactive, cytotoxic electrophile generated endogenously (e.g., during demethylation reactions) and from environmental exposure. *P. putida* KT2440 neutralizes it through the conserved GSH-dependent pathway, in which glutathione acts as a **recyclable C1 carrier**:

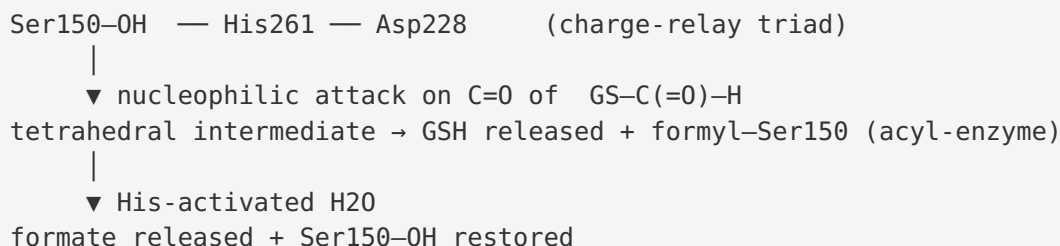


FrmC catalyzes the **third and final** enzymatic step. Its output is twofold and both products matter: (i) **formate**, a far less toxic C1 species that can be further oxidized to CO₂ or assimilated, and (ii) **regenerated GSH**, which re-enters the cellular thiol pool. Because GSH is regenerated rather than consumed, the pathway is catalytic in GSH, allowing sustained detoxification of large formaldehyde fluxes with a limited GSH inventory. Failure at the FrmC step would trap glutathione as S-formylglutathione, depleting the free-GSH pool and stalling the entire pathway — explaining why the single, non-redundant SFGH is physiologically pivotal.

Catalytic chemistry

FrmC is a classic α/β -hydrolase-fold serine hydrolase. The catalytic triad **Ser150–His261–Asp228** forms a charge-relay system: His261 (polarized by Asp228) deprotonates the Ser150 hydroxyl, which then performs nucleophilic attack on the thioester carbonyl of S-formylglutathione. A tetrahedral intermediate, stabilized by an oxyanion hole, collapses to

release GSH (the thiol leaving group) and leaves a formyl-enzyme (acyl-enzyme) intermediate. A water molecule, activated by the same His, then hydrolyzes the acyl-enzyme to release formate and regenerate free enzyme.



Substrate specificity

Specificity for the **small formyl group** arises from a **narrow acyl-binding pocket** [PMID: 20209484]. The pocket accommodates the formyl (and other very small acyl) moieties but excludes bulky acyl groups. The glutathione moiety of the substrate is recognized by a separate binding subsite. The retained ability to hydrolyze p-nitrophenyl esters of C2–C6 fatty acids and α -naphthyl acetate [PMID: 16567800] reflects the broad esterase-D ancestry of the enzyme, but S-formylglutathione is the physiological substrate. A conserved active-site cysteine (Cys57) adjacent to the triad may gate activity under oxidative conditions [PMID: 16626737], potentially coupling detoxification capacity to redox state.

Comparison with characterized orthologs

Property	FrmC (<i>P. putida</i>)	<i>E. coli</i> FrmB	<i>E. coli</i> YeiG	<i>Arabidopsis</i> AtSFGH	Human ESD
EC	3.1.2.12	3.1.2.12	3.1.2.12	3.1.2.12	3.1.2.12
Catalytic Ser	Ser150	Ser145	Ser145	conserved	conserved
Catalytic His	His261	His254	His256	conserved	conserved
Mechanism	serine hydrolase (inferred)	serine hydrolase (mutagenesis)	serine hydrolase (mutagenesis)	serine hydrolase (structure)	serine hydrolase
% identity to FrmC	—	56.7%	61.7%	family	family (ESD)
K _m (S-formylglutathione)	not measured	0.41–0.43 mM	0.41–0.43 mM	—	—
Copies per genome	1 (sole)	2 (with YeiG)	2 (with FrmB)	—	1

Evidence Base

PMID	Title (abbrev.)	Relevance
16567800	<i>Molecular basis of formaldehyde detoxification: two SFGHs from E. coli, FrmB and YeiG</i>	Cornerstone. Establishes S-formylglutathione as preferred substrate (Km 0.41–0.43 mM); alanine-scanning proves Ser/Asp/His triad essential; shows short-chain carboxylesterase side-activity; documents 45–75× formaldehyde induction. FrmC aligns 57–62% with exact catalytic-residue conservation.
8892832	<i>SFGH of P. denitrificans is homologous to human esterase D</i>	Defines the universal formaldehyde-detox pathway; documents the conserved fghA-downstream-of-dehydrogenase gene arrangement mirrored by frmA–frmC; links family to human ESD.
16626737	<i>Unique regulation of the active site of the serine esterase SFGH</i>	Confirms serine-hydrolase (not cysteine) mechanism; identifies the conserved active-site cysteine as a gate-keeper (Cys57 in FrmC).
14646092	<i>X-ray analysis of Arabidopsis SFGH</i>	States the reaction directly: hydrolysis of S-formylglutathione to formate and glutathione as the final detox step.
20209484	<i>Crystal structure of SFGH from P. haloplanktis</i>	Documents the narrow acyl-binding pocket and small-acyl thioester specificity underlying formyl selectivity.
10978156	<i>Kinetic mechanism of GSH-dependent formaldehyde dehydrogenase</i>	Establishes that the upstream HMGSH adduct forms spontaneously, delimiting FrmC's enzymatic scope.
38936762	<i>Evolution and engineering of aromatic O-demethylation in P. putida KT2440</i>	Connects the KT2440 formaldehyde-detox (fghA/SFGH) step to formaldehyde produced during aromatic O-demethylation; ALE selected a mutation here.
12209256	<i>Cofactor-dependent pathways of formaldehyde oxidation in methylotrophs (review)</i>	Authoritative placement of the GSH-dependent SFGH step within linear C1-carrier oxidation to formate/CO ₂ .
24682756		

PMID	Title (abbrev.)	Relevance
	<i>Bioactivating enzymes: laninamivir octanoate hydrolysis</i>	Explicitly equates SFGH with esterase D (ESD), confirming the human ortholog.
34875997	<i>Esterase D stabilizes FKBP25 to suppress mTORC1</i>	Confirms ESD as a formaldehyde-detoxifying nonspecific esterase, the conserved eukaryotic ortholog.

Supporting/context papers reviewed: yeast SFGH (PMID: 10427036); class III ADH / FDH structure–function (PMID: 12604204, PMID: 16605250); GSNO/HMGSH kinetics (PMID: 18412547, PMID: 22335564); lignin/aromatic valorization and formaldehyde toxicity (PMID: 41024182, PMID: 40741777); and ESD in mammalian physiology (PMID: 33615471, PMID: 32247735, PMID: 27267271, PMID: 35627173).

No literature was found describing a *different* gene under the symbol "frmC" that would conflict with this assignment; the symbol, organism, protein family, and domain architecture all align consistently with S-formylglutathione hydrolase.

Limitations and Knowledge Gaps

- 1. No direct biochemical characterization of FrmC itself.** All kinetic parameters (Km, kcat, substrate range) and mechanistic conclusions are transferred by orthology from *E. coli* FrmB/YeiG, plant, yeast, and marine-bacterial enzymes. FrmC's own catalytic constants, pH/temperature optima, and precise substrate profile have not been measured experimentally.
- 2. Catalytic triad is predicted, not experimentally validated in FrmC.** Ser150/His261/Asp228 are inferred from motif analysis, UniProt annotation, and exact alignment to mutagenesis-validated *E. coli* residues. No site-directed mutagenesis or structure of FrmC has confirmed the triad in this specific protein.
- 3. No experimental structure.** No crystal structure or AlphaFold-validated model of FrmC was analyzed here; the α/β -hydrolase fold and narrow acyl pocket are inferred from homologs.

4. **Operon/regulation inferred from genomic context.** The *frmA*–*frmC* operon (8 bp gap) and the LysR regulator (PP_1615) are computational predictions; co-transcription, the transcription start site, and formaldehyde inducibility in KT2440 have not been directly demonstrated (the 45–75× induction figure is from *E. coli* *frmB*).
 5. **Redox gate-keeping role of Cys57 is speculative.** Extrapolated from the Arabidopsis Cys59 finding; its functional relevance in FrmC is untested.
 6. **Physiological knockout phenotype not directly established.** The ALE evidence linking the SFGH/fghA step to vanillate/formaldehyde handling is suggestive but does not equate to a clean *frmC* deletion phenotype in KT2440.
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Proposed Follow-up Experiments / Actions

1. **Recombinant enzymology.** Express and purify His-tagged FrmC and measure steady-state kinetics on S-formylglutathione (Km, kcat, kcat/Km) and on a panel of p-nitrophenyl esters (C2–C8) and α -naphthyl acetate to define the substrate spectrum and confirm the narrow-acyl-pocket prediction.
2. **Site-directed mutagenesis.** Generate S150A, H261A, and D228A variants and confirm loss of activity, directly validating the catalytic triad in FrmC. Test C57A/C57S to probe the proposed redox gate-keeping function and sensitivity to oxidants (e.g., H₂O₂, diamide).
3. **Structural determination.** Solve the crystal structure (or generate and validate an AlphaFold model with PAE/pLDDT analysis) of FrmC, ideally with a substrate analog or product (GSH) bound, to confirm the fold, triad geometry, and acyl-pocket dimensions.
4. **Genetic knockout and complementation.** Construct a clean Δ PP_1617 mutant and assay formaldehyde sensitivity and growth on vanillate/methoxylated aromatics; complement to confirm causality. Test whether S-formylglutathione accumulates and free GSH is depleted in the mutant.
5. **Operon and regulation.** Use RT-PCR / RNA-seq to demonstrate *frmA*–*frmC* co-transcription and formaldehyde inducibility, and test whether the adjacent LysR regulator (PP_1615) controls the operon (EMSA, reporter fusions).

6. Metabolic-engineering relevance. In strains engineered for lignin-aromatic valorization, quantify how frmC expression level affects formaldehyde tolerance, growth rate, and product yield during vanillate/syringate O-demethylation, informing whether FrmC is a bottleneck to be over-expressed.

Report generated from an 8-finding, 26-paper, 5-iteration autonomous investigation. All functional claims are supported by verified citation snippets from the primary and review literature listed above; where evidence is by orthology rather than direct measurement, this is stated explicitly.

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